IMPLEMENTION OF A FUZZY INFERENCE SYSTEM

Aim:

To implement Fuzzy Inference System.

PROGRAM:

```
import numpy as np
import skfuzzy as fuzz
import matplotlib.pyplot as plt

temperature = np.arange(0, 41, 1)
fan_speed = np.arange(0, 11, 1)

temp_low = fuzz.trimf(temperature, [0, 0, 20])
temp_medium = fuzz.trimf(temperature, [10, 20, 30])
temp_high = fuzz.trimf(temperature, [20, 40, 40])

fan_low = fuzz.trimf(fan_speed, [0, 0, 5])
fan_medium = fuzz.trimf(fan_speed, [2, 5, 8])
fan_high = fuzz.trimf(fan_speed, [5, 10, 10])

plt.figure(figsize=(10, 6))

plt.subplot(2, 1, 1)
plt.plot(temperature, temp_low, label="Low")
plt.plot(temperature, temp_medium, label="Medium")
plt.plot(temperature, temp_high, label="High")
plt.title("Temperature Membership Functions")
```

```
plt.xlabel("Temperature (C)")
plt.ylabel("Membership Degree")
plt.legend()

plt.subplot(2, 1, 2)
plt.plot(fan_speed, fan_low, label="Low")
plt.plot(fan_speed, fan_medium, label="Medium")
plt.plot(fan_speed, fan_high, label="High")
plt.title("Fan Speed Membership Functions")
plt.xlabel("Fan Speed")
plt.ylabel("Membership Degree")
plt.legend()

plt.tight_layout()
plt.show()

temperature_input = 28

temp_low_level = fuzz.interp_membership(
    temperature, temp_low, temperature_input
)
temp_medium_level = fuzz.interp_membership(
    temperature, temp_medium, temperature_input
)
temp_high_level = fuzz.interp_membership(
    temperature, temp_high, temperature_input
)

low_rule = temp_low_level
medium_rule = temp_medium_level
```

```
high_rule = temp_high_level

activated_low = np.fmin(low_rule, fan_low)
activated_medium = np.fmin(medium_rule, fan_medium)
activated_high = np.fmin(high_rule, fan_high)

aggregated = np.fmax(
    activated_low,
    np.fmax(activated_medium, activated_high)
)

fan_output = fuzz.defuzz(fan_speed, aggregated, "centroid")

print("Temperature:", temperature_input, "C")
print("Fuzzified fan speed:", round(fan_output, 2))
```

OUTPUT:

Temperature: 28 C
Fuzzified fan speed: 6.00